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Ajinkya J. Naik

Education

2024–2025 **MASt in Theoretical Physics (Part III Mathematics)**, *University of Cambridge*
Essay/Thesis: Proposals for Tabletop Tests of Quantum Gravity (Distinction, 87/100)

2020–2023 **B.Sc. in Physics**, *MES Abasaheb Garware College, University of Pune*
Subjects: Physics (specialization), Statistics, Mathematics, Electronics
Thesis: Detecting H-I Emissions from the Galactic Plane of Milky Way (Grade A+)

Work Experience

2026–Present **Project Assistant**, *Inter-University Centre for Astronomy & Astrophysics (IUCAA)*

- Developed experiments for astronomy education and research such as solar photometry, two-element radio interferometer for solar observations (in progress, leading), antenna characterisation, and H-I line emission mapping of the Milky Way.
- Coordinated and attended on-site schools and workshops, including selecting participants and conducting tutorial sessions.
- Designed and maintain a website hosting the centre's digital learning resources.
- Involved in teaching and outreach events like the annual National Science Day program.

Research Experience

2026–Present **Research Collaboration**, *Majumdar group, NISER, Bhubaneswar*

- Building a joint atmosphere-interior evolution framework for sub-Neptunes and terrestrial planets: I developed the structural solver and evolution model (adiabatic interior, flexible layering from core to radiative atmosphere), validated against the observed masses and radii of sub-Neptunes and terrestrial planets.
- Coordinating and leading an undergraduate intern building the global chemical equilibrium module.

2025–2026 **Cambridge Mathematics Placement**, *Handley Lab, Institute of Astronomy, Cambridge*

- Built **NestAR**, a GPU-accelerated nested sampling framework in Python/`JAX` for Bayesian ARIMA model selection in astronomical time series; released as a documented, installable Python package (https://github.com/najinkya1313/NestAR).
- Applied it to sunspot number forecasting and to photometric variability in Kepler and TESS light curves.
- Mitigated correlated stellar-rotation noise in the Kepler-17 light curve while preserving the Kepler-17b transit signal in the residuals.
- First-author paper under revision at *RAS Techniques and Instruments* (https://arxiv.org/abs/2512.01929); presented at the NASA Sagan Summer Workshop (poster).

2024–2025 **Part III Essay**, *DAMTP, University of Cambridge*

- Reviewed experimental proposals for testing the quantum nature of gravity (BMV protocol, non-Gaussianity tests in BECs) and proposed a geometrical optimisation to shorten the experiment timescale.

2023–2024 **Project Student**, *Dr. Isobel Romero-Shaw, University of Cambridge*

- Used the population synthesis code COMPAS to model binary black holes from high-mass X-ray binaries and fit their effective-spin distribution with Gaussian mixture models to be used as informative priors for a hierarchical Bayesian analysis of LIGO population data.

2022–2024 **Bachelor's Thesis**, *MES Abasaheb Garware College, University of Pune and IUCAA*

- Observed the 21 cm line from the Galactic plane with a horn antenna at IUCAA's Radio Physics Lab; characterised the radiometer and measured line-of-sight velocities from Doppler shifts in Python.

Short Projects

2023 **GRB 210306A : A Multi-Wavelength Analysis**, *ARIES, Nainital*, Sep–Nov

2022 **Generating Post-Newtonian gravitational waveform templates**, *IIT Bombay*, Jul–Sep

Papers

2025 A. Naik and W. Handley, *Nested Sampling for ARIMA Model Selection in Astronomical Time Series (*under revision at RAS Techniques and Instrumentation)*, http://arxiv.org/abs/2512.01929.

Workshops and Schools

Nov 2026 **Course on Solar Physics and Planetary Science**, *Physical Research Laboratory/CSSTEAP*

Aug 2026 **Statistical Techniques in Astronomy**, *IUCAA*

Jul 2026 **Sagan Summer Workshop**, *NexSci (NASA)*

Jul 2023 **Summer School on Gravitational Wave Astronomy**, *ICTS-TIFR*

Dec 2022 **15th Radio Astronomy Winter School**, *NCRA-TIFR and IUCAA*

Talks and Presentations

2026 **NASA Sagan Summer Workshop**, *NexSci-Caltech*

Remote poster presentation on "Disentangling Exoplanet Transits with NestAR" (https://nexsci.caltech.edu/workshop/2026/posters/Poster_AjinkyaNaik_16.pdf).

2025 **Cambridge Mathematics Placement Presentation Day**, *University of Cambridge*

"Nested Sampling for ARIMA Model Selection – A Novel Approach to Astronomical Time Series Analysis".

2025 **Part III Seminar Series**, *DAMTP, University of Cambridge*

"Surfing The Waves of Spacetime: An Introduction to Gravitational Wave Theory and Detection".

2023 **Visiting Student Seminar**, *ARIES, Nainital*

"GRB 210306A: A Multi-Wavelength Analysis".

Scholarships, Awards and Grants

2026 **Sir Ratan Tata Trust Award**, *6,75,000 INR*

*Awarded for academic performance and research during master's program.

2025 **IoA Summer Projects Fund**, *3,800 GBP*

2024 **Oxford Cambridge Society of India Anita Banerji Memorial Scholarship**, *4,00,000 INR*

2024 **JN Tata Endowment and Travel Grant**, *11,00,000 INR*

2024 **Narotam Sekhsaria Postgraduate Scholarship**, *15,00,000 INR*